ZACKARY HORTON

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Education

Doctor of Philosophy: September 2025-Present
McGill University – Montréal, QC
- Research focused degree studying algebraic number theory and automorphic forms
- Supervised by Professor Patrick Allen, Professor Henri Darmon

Master of Science: Pure Mathematics, September 2024-September 2025
Imperial College London - London, UK
- One year of full-time study along with a master's thesis project
- Studied topics and took classes necessary to continue deeper into research

Bachelor of Science: Honours Mathematics, Graduated May 2024
McGill University - Montréal, QC
- CGPA: 3.70/4.0
- Two-time Dean's list recipient
- Awarded honours with distinction upon graduating

Research experience

Master's thesis project: Summer 2025
Imperial College London
- Studied local class field theory through a Galois cohomological lens
- Worked under the supervision of Professor Alexei Skorobogatov

Undergraduate summer research: Summer 2023
McGill University
- Awarded grant from the Natural Sciences and Engineering Research Council of Canada
- Studied modular forms and Serre's strong modularity conjecture
- Used SageMath to compute instances of non-minimal nebentypus
- Worked efficiently in a team of three people under the supervision of Professor Patrick Allen

Undergraduate Project: Winter 2024
McGill University
- Studied Quaternion Algebras by Voight to learn about tensors, algebras, maximal orders and connecting ideals
- Used this to produce a program that computed the graphs of maximal orders with connecting ideals as edges

Extracurricular and other experience

Teaching Assistant: January 2026-Present
McGill University
- Practiced developing tutorials and problems to help with students' understanding
- Filled in for lectures when needed

Math Tutor, McGill Math Help Desk: 2023
McGill University
- Improved ability to communicate mathematical concepts
- Shared passion for math with others

Airtanker Base Loader and Supervisor: Summers 2020- 2022, 2024
GNWT
- Supervised employees and maintained air bases across the Northwest Territories
- Travelled across the territory to follow forest fire danger and mitigate risk of structure damage

Team Member, Assistant Coach: 2019-Present
McGill Volleyball Team
- Learned to thrive in competitive environment
- Developed team building and leadership skills

Skills

- Language Skills: Native English, intermediate French
- Computer Skills: Proficiency with Python, SageMath, Java, C, and Unix-like systems for storage and computation